GRADE 7 ACCELERATED · UNIT 6 · LESSON 4

# Pythagorean Theorem in Real-World Problems

*Independent Set 6.04* **ANSWER KEY**

Name
Date
Period

*Six questions, on your own, in silence. Show the step that does the work.*

**1.** A tablet screen is 5 inches wide and 8 inches tall. How long is its diagonal, to the nearest tenth of an inch?

***About 9.4 in***

*25 + 64 = 89; √89 ≈ 9.43.*

**2.** A fire truck's 50-ft ladder reaches 48 ft up a building. How far from the building is its foot?

***14 ft***

*2,500 − 2,304 = 196; the ladder is c.*

**3.** A ramp rises 3 ft over a run of 14 ft. How long is the ramp, to the nearest tenth of a foot?

***About 14.3 ft***

*9 + 196 = 205; √205 ≈ 14.32.*

**4.** A park is 7 m by 11 m. How much shorter is the walk straight across, corner to corner, than the walk along two sides? Give it to the nearest tenth of a meter.

***About 5.0 m shorter***

*Along two sides: 18 m. Across: √170 ≈ 13.04 m. 18 − 13.04 = 4.96, so 5.0.*

**5.** Will a 10-in pencil lie flat in a box 7 inches by 8 inches, corner to corner?

***Yes — the diagonal is about 10.6 in.***

*49 + 64 = 113; √113 ≈ 10.63 > 10.*

**6.** A kite flies on 19 m of tight string. Straight below the kite, level with the hand that holds the string, is a point 12 m from the hand. How high is the kite above that point, to the nearest tenth of a meter?

***About 14.7 m***

*361 − 144 = 217; √217 ≈ 14.73; the string is c.*